# Hand Derivation 1: Liquidity Shortfall and Surplus

## 1. Definitions

$$w_K = \beta_K A + \varepsilon_K$$
$$S_K = d_K (w_K + \mu_K), \quad d_K > 0$$
$$D_K(A, \mu_K) = E[(-S_K)^+ | A]$$
$$X_K(A, \mu_K) = E[(S_K)^+ | A]$$

Since deposits are fixed and positive:

$$D_K / d_K = E[(-w_K - \mu_K)^+ | A]$$
$$X_K / d_K = E[(w_K + \mu_K)^+ | A]$$

Assume integrable shocks and an atomless conditional distribution

## 2. Shortfall derivative

Define

$$L_K(\mu_K, A) = \frac{D_K(A, \mu_K)}{d_K}$$

For a performance beyond the breaking point:

$$\frac{\partial}{\partial \mu_K}(-w_K - v_K)^+ = -1_{\{w_K + \mu_K < 0\}}$$

So

$$L_{K,\mu} = E[-1_{\{w_K + \mu_K < 0\}} | A]$$
$$= -\Pr(w_K + \mu_K < 0 | A)$$

Define

$$\pi_K^-(\mu_K, A) = \Pr(w_K + \mu_K < 0 | A)$$
$$L_{K,\mu} = -\pi_K^-$$

## 3. Surplus derivative

Now define

$$M_K(\mu_K, A) = \frac{X_K(A, \mu_K)}{d_K}$$

Derivate

$$\frac{\partial}{\partial \mu_K}(w_K + \mu_K)^+ = 1_{\{w_K + \mu_K > 0\}}$$

Therefore

$$M_{K,\mu} = E[1_{\{w_K + \mu_K > 0\}} | A]$$
$$= \Pr(w_K + \mu_K > 0 | A)$$

Define

$$\pi_K^+(\mu_K, A) = \Pr(w_K + \mu_K > 0 | A)$$
$$M_{K,\mu} = \pi_K^+$$

The derivative and expectation can be interchanged because the positive-part function is Lipschitz and the conditional shock distribution has no atom at the threshold

## 4. Human verification: ratio versus amount

$$D_K = d_K L_K$$

As $d_K$ is fixed

$$D_{K,\mu} = \partial(d_K L_K) / \partial \mu_K$$
$$= -d_K \pi_K^-$$

Analogously

$$\chi_{K,M} = \partial_K \pi_K^+$$

5. Numerical unit check

$$\partial_K = 1.6, \quad \pi_K^- = 0.3$$

So

$$L_{K,M} = -0.3$$
$$D_{K,M} = (-1.6)(0.3)$$
$$D_{K,M} = -0.48 \neq -0.3$$

Human verdict: the probability derivatives applies directly to the normalized shortfall. The monetary shortfall derivative must include the deposit factor